The Faculty of Medicine of Harvard University
Curriculum Vitae

Date Prepared:	**October 7th, 2026**
Name:	**Xiang Li**
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Place of Birth:	**China**

Education:

09/2006	B.E.	Automation	Shanghai Jiaotong University, China
06/2016	PhD	Computer Science	University of Georgia

Postdoctoral Training:

08/2016-07/2019	Research Fellow	Radiology	Harvard Medical School and Massachusetts General Hospital

Faculty Academic Appointments:

08/2019-05/2023	Instructor	Radiology	Harvard Medical School
06/2023-	Assistant Professor	Radiology	Harvard Medical School
09/2023-	Affiliate Faculty Member	Kempner Institute for Natural and Artificial Intelligence	Harvard University

Appointments at Hospitals/Affiliated Institutions:

08/2019-	Research Staff	Radiology	Massachusetts General Hospital

Major Administrative Leadership Positions:

International

2019, 2022-2024	Founding program chair for the *International Workshop on Multiscale Multimodal Medical Imaging*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2021	Organizer and program chair for the *International Workshop on Multimodal Learning and Fusion Across Scales for Clinical Decision Support*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2024	Organizer and program chair for the *International Workshop on Embodied AI and Robotics for HealTHcare*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)

2024	Organizer and program chair for the *SAGES Critical View of Safety Challenge*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2024	Organizer and program chair for the *NeurIPS Workshop on Advancements in Medical Foundation Models*	NeurIPS Foundation
2025	Organizer and program chair for the *AAAI Workshop: AI for Health: Leveraging Artificial Intelligence to Revolutionize Healthcare*	The Association for the Advancement of Artificial Intelligence (AAAI)
2025	Organizer and program chair for the *ISBI Special Session: Unlocking the Potential of Foundation Models in Biomedicine: Opportunities and Challenges*	IEEE Engineering in Medicine and Biology Society (EMBS)
2026	Organizer and program chair for the *SIGGRAPH Frontiers* in *Quantum Machine Learning: Bridging Quantum Computing & Machine Learning*	Association for Computing Machinery (ACM)

Committee Service:

International

2019	Program Committee	NeurIPS Workshop on Machine Learning in Computational Biology
2019-2022	Program Committee	ACM SIGKDD Workshop on Mining and Learning from Time Series
2023	Program Committee	International Workshop on Medical Image Learning with Noisy and Limited Data
2023-2024	Area Chair	International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)
2025	Program Committee	IEEE Symposium on Computer-Based Medical Systems
2025-	Area Chair	Conference on Neural Information Processing Systems (NeurIPS)
2025-	Area Chair	Association for the Advancement of Artificial Intelligence (AAAI) Conference
2025-	Area Chair	Association for Computing Machinery's Special Interest Group on Knowledge Discovery and Data Mining (KDD)
2026	Area Chair	Association for Computing Machinery International Conference on Multimedia
2026	Committee	SNMMI Freek J. Beekman PIDSC Young Investigator Award

Professional Societies:

2011-	Institute of Electrical and Electronics Engineers (IEEE), International	Member
2011-	Engineering in Medicine and Biology Society (EMBS), International	Member
2011-	The Medical Image Computing and Computer-Assisted Intervention Society, International (MICCAI), International	Member
2023-	American Medical Informatics Association (AMIA), National	Member
2023-	Society of Nuclear Medicine and Molecular Imaging (SNMMI), International	Member

Grant Review Activities:

2024	ZRG1 CTH-E (57): PAR Panel: Academic-Industrial Partnerships for Translation of Technologies 10/16/2024	National Institutes of Health, Division of Translational and Clinical Sciences (DTCS) Ad hoc reviewer
2025	ZMH1 ERB-E (03) S: BRAIN Initiative: Research on the Ethical Implications of Advancements in Neurotechnology and Brain Science 05/02/2024	National Institutes of Health, BRAIN Initiative Ad hoc reviewer
2026	Bradley-Alavi Student Fellowship	Society of Nuclear Medicine and Molecular Imaging (SNMMI) Study section member
2026	Round Open Technology Programme OTP	Dutch Research Council (NWO) Domain Applied and Engineering Sciences (AES) Ad hoc reviewer

Editorial Activities:

- **Ad hoc Reviewer**

IEEE Transactions on Pattern Analysis and Machine Intelligence

JAMA Network Open

Nature Human Behavior

Nature Medicine

Nature Biomedical Engineering

- **Other Editorial Roles**

2021-	Associate Editor	Frontiers in Oncology
2021-	Associate Editor	Frontiers in Radiology
2021-	Associate Editor	Frontiers in Neuroscience
2021	Guest Editor	Frontiers in Neuroscience, Special Issue "Multi-Dimensional Characterization of Neuropsychiatric Disorders"

2022-	Associate Editor	Frontiers in Cardiovascular Medicine
2023-	Associate Editor	Meta-Radiology
2023-	Associate Editor	BMC Biomedical Engineering
2023-	Associate Editor	IEEE Transactions on Artificial Intelligence
2024-	Associate Editor	Data Intelligence
2024	Guest Editor	IEEE Transactions on Neural Networks and Learning Systems, Special Issue "Advancements in Foundation Models"
2024-	Associate Editor	Connected Health And Telemedicine
2024-	Associate Editor	BMC Artificial Intelligence
2025-	Guest Associate Editor	IEEE Transactions on Medical Imaging
2025	Associate Editor	Scientific Report
2025	Ad-hoc Editor	Proceedings of the National Academy of Sciences of the United States of America
2026	Guest Editor	Neurocomputing "Recent Advancements in Foundation Models: Reviews and Surveys"

Honors and Prizes:

2011	Best Student Paper Award	IEEE International Symposium on Biomedical Imaging	"Brain State Change Detection via Fiber-centered Functional Connectivity Analysis"
2013	Best Student Paper Award	IEEE International Symposium on Biomedical Imaging	"Discovering Common Functional Connectomics Signatures"
2015	Paul D. Coverdell Neuroimaging Franklin Foundation Scholars Program Travel Award	Department of Psychology, University of Georgia	
2015	Cover and Feature Paper	IEEE Transactions on Biomedical Engineering	"Holistic atlases of functional networks and interactions reveal reciprocal organizational architecture of cortical function"
2016	Outstanding Graduate Dissertation/Thesis	University of Georgia	
2018	Most Cited Articles	Journal of the American College of Radiology	"Artificial Intelligence and Machine Learning in Radiology: Opportunities, Challenges, Pitfalls, and Criteria for Success"
2020	Best Paper Awards	IEEE International Symposium on Biomedical Imaging	"ASCNet: Adaptive-Scale Convolutional Neural Networks for Multi-Scale Feature Learning"
2021	MGH Thrall Innovation Grants Award	Massachusetts General Hospital	"Chest Radiographs-based Lung Cancer Screening by the DeepProjection Technique"

2022	Best Paper Awards	IEEE Transactions on Radiation and Plasma Medical Sciences	"Deep Learning-Based Image Segmentation on Multimodal Medical Imaging"
2024	Research Scholar Program	Google Inc.	"Tailoring Large Language Models for the Diagnosis and Management of Late-life Depression Patients with Limited Access to Healthcare Resources"
2024	Distinguished Paper Award	American Medical Informatics Association (AMIA)	"MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering"
2025	Featured Article	IEEE Reviews in Biomedical Engineering	"Artificial General Intelligence for Medical Imaging Analysis"
2025	Editor's Choice Article	Medical Physics	"Fine-tuning open-source large language models to improve their performance on radiation oncology tasks: A feasibility study to investigate their potential clinical applications in radiation oncology"

Report of Funded and Unfunded Projects

Past

2022-2025	"Identification of Multi-modal Imaging Biomarkers for Early Prediction of MCI-AD Conversion via Multigraph Representation" NIH R03, 1R03AG078625-01, Direct cost: $200,000 /2 years **Principal Investigator** To investigate the interaction among structural, functional, and proteinopathies networks in MCI and AD patients to identify and evaluate image biomarker for AD conversion.
2024-2025	"Tailoring Large Language Models for the Diagnosis and Management of Late-life Depression Patients with Limited Access to Healthcare Resources" Google Inc., Direct cost: $100,000 /1 year **Principal Investigator** To explore the feasibility of using LLMs for managing late-life depression patients, especially on its differential diagnosis with depression, and assistant to the caregivers.
2024-2026	"Simulated Computed Tomography Scans Generated from X-Rays Using a Diffusion Model" American Foundation for Surgery of the Hand (AFSH), Direct cost: $60,000 /2 years **MPI** To develop an AI model that can generate near-real wrist CTs from X-ray images based on the conditional diffusion model.

Current

2026-2030	"360° Collaborative Hub for AI in TMD Research (360° Chat-TMD)" NIH U54, 1U54DE035412-01, Direct cost: $5,584,812 /5 years **Co-Investigator** To establish a Center of Excellence in TMD Research, and Training and Outreach.
2026-2029	"UniBrain: An End-to-End Unified Toolkit for Comprehensive Neuroimaging Analysis" NIH R01, 1R01EB038264-01A1, Direct cost: $2,525,515 /4 years **MPI** To develop AI-empowered technologies and solutions for neuroimaging analysis.
2026-2029	"Automated PET Report Generation with Visual Grounding" NIH R01, 1R01EB038426-01, Direct cost: $2,304,244 /4 years **Subcontract PI (MGH)** To develop visually grounded findings through a PET report generation model using federated learning.

Projects Submitted for Funding

2026	"Personalized Modeling of Amyloid-β/Tau Causal Relationship to Inform Emerging Treatments for Alzheimer's Disease" NIH R01, Direct cost: $3,311,165/5 years Principal Investigator Identification of regional-specific causal relationships between amyloid-β and tau in Alzheimer's disease patients by PET imaging using advanced machine learning approaches.

Report of Local Teaching and Training

Research Supervisory and Training Responsibilities:

2018-	Supervised graduate research interns and research fellows (average 2-3 per year).	Massachusetts General Hospital. Two hours of lab meetings per week; One hour per week per student and research fellow for 1:1 discussion and supervision.
2021-	Administration and coordination of the CAMCA AIxMedicine Biweekly Seminar, the public lab meeting on AI research in medicine of the Center for Advanced Medical Computing and Analysis in MGH.	Massachusetts General Hospital. One hour of meetings per two weeks.

Other Mentored Trainees and Faculty:

2019-2023	Hui Ren, Research Fellow at MGH. **Mentoring role**: Advising research on cardiac image analysis and Electronic Health Record (EHR) modeling.
2019-2020	Haixing Dai, Ph.D. student at the University of Georgia Graduate research intern at MGH. **Mentoring role**: Advising research on graph-based deep learning models for brain network modeling from MEG and PET images.
2020	Qinglin Dong, Research Fellow at MGH. **Mentoring role**: Advising research on neuroimage, especially functional MRI analysis and modeling.
2021-2023	Jerome Charton, Research Fellow at MGH. **Mentoring role**: Advising research on the computational modeling of ultrasound, especially echocardiogram images.
2022-2023	Zhengliang Liu, Ph.D. student at the University of Georgia Graduate research intern at MGH. **Mentoring role:** Advising research on natural language processing (NLP) in healthcare.

2022-2025	Zhennong Chen, Research Fellow at MGH. **Mentoring role:** Advising research on the image analysis of cardiac MRI.
2022-2025	Sekeun Kim, Research Fellow at MGH. **Mentoring role:** Advising research on the differential diagnosis for cardiac conditions and image modeling for echocardiography.
2023-2025	Siyeop Yoon, Research Fellow at MGH. **Mentoring role:** Advising research on the generative modeling for medical images utilizing diffusion models.

Local Invited Presentations:

☒ *No presentations below were sponsored by 3rd parties/outside entities*

2025 "Engineering Large Pre-trained Models: Techniques and Best Practices" (**Invited Talk**)
School of Engineering and Applied Sciences, Harvard University, Cambridge, MA

2026 "Generative AI for Healthcare: Alignment, Foundation Models, Simulation, and Agentic Systems towards Tangible Clinical Intelligence" (**Invited Talk**)
Biogen/Sanofi Seminar Series, Boston, MA

Report of Regional, National and International Invited Teaching and Presentations

☒ *No presentations below were sponsored by 3rd parties/outside entities*

Regional

2024 "Utilizing Large Language Models for Effective Management and Analysis of Nuclear Medicine Radiology Reports" (**Invited Talk**)
Hampton Symposium 2024, Boston, MA

National

2017 "Towards Practical Problems in Deep Learning for Radiology Image Analysis" (**Invited Talk**)
Nvidia GPU Technology Conference, San Jose, CA.

2018 "Deep Learning Algorithm for rapid automatic detection of pneumothorax on chest CT" (**Selected Oral Abstract**)
Annual Meeting of American Roentgen Ray Society, Washington, D.C.

2019 "Personalized Healthcare for Heart Failure with Preserved Ejection Fraction (HFpEF): Diagnosis, Phenotyping and Treatment Optimization with Imaging and EHR Data" (**Invited Talk**)
Department of Statistics, The University of Georgia, Athens, GA.

2022 "Novel Methodologies for Combined Image and EMR Modeling" (**Invited Talk**)
Department of Computer Science and Engineering, The University of Texas at Arlington, Arlington, TX

2022 "Data Governance of the SAGES CVS Challenge" (**Invited Talk**)
Society of American Gastrointestinal and Endoscopic Surgeons, Houston, TX (Virtual)

2023 "Application and Development of Foundational Models in Healthcare" (**Invited Talk**)
Department of Electrical and Computer Engineering, University of Rochester, Rochester, NY (Virtual)

2023 "Application and Development of Foundation Models in Healthcare" (**Invited Talk**)
AIM Seminar Series, Department of Radiation Oncology, UT Southwestern Medical Center, Dallas, TX.

2023 "A New Perspective of Human-Computer Interaction in the Era of Large Pre-trained Models" (**Keynote Speech**)

	Workshop on the Intersection of Artificial Intelligence and Human Intelligence (IAIHI), Hoboken, NJ.
2023	"Application and Development of Foundation Models in Healthcare" (**Invited Talk**) *Neuro Image Research and Analysis Laboratories (NIRAL)*, Department of Psychiatry, University of North Carolina at Chapel Hill, Chapel Hill, NC.
2024	"Development of Specialized Large Language Models for Radiology Report Processing" (**Keynote Speech**) *31st Annual Council on Ionizing Radiation Measurements & Standards Meeting*, Rockville, MD.
2024	"Understanding Neurodegenerative Diseases via Multi-modal Data Analytics, Generative Modeling, and Counterfactual Causal Inference" (**Invited Talk**) *The New Investigators in Alzheimer's Disease Meeting*, Bethesda, MA
2025	"Generative AI in Healthcare: Foundation, Practice, and Perspectives" (**Invited Talk**) *Department of Computer Science*, Stony Brook University, New York, NY
2026	"Flash Quantum Surface-Code Decoding on GPU Thread-Block Clusters" (**Invited Talk**) *Department of Physics & Astronomy*, University of Tennessee, Knoxville, TN

International

2019	"Automated Segmentation of Cervical Nuclei in Pap Smear Images using Deformable Multi-path Ensemble Model" (**Selected Oral Full-length Paper**) *IEEE International Symposium on Biomedical Imaging*, Venice, Italy.
2023	"Application and Development of Foundational Models in Healthcare" (**Invited Talk**) *IEEE EMBS Webinar Series* (Virtual)
2023	"Identification of Causal Relationship between Amyloid-beta Accumulation and Alzheimer's Disease Progression via Counterfactual Inference" (**Selected Oral Abstract**) *Optica Imaging Congress*, Boston, MA
2023	"Recent Advances in Sparse and Ultra-Sparse Reconstruction and Generative Modeling for Medical Imaging" (**Invited Talk**) *Optica Imaging Congress*, Boston, MA
2024	"Cross-Modal Retrieval for Alzheimer's Disease Diagnosis Using CLIP-Enhanced Dual Deep Hashing" (**Selected Oral Abstract**) and "Conditional Score-Based Diffusion Model for Cortical Thickness Trajectory Prediction" (**Selected Oral Abstract**) *AMIA 2024 Annual Symposium*, San Francisco, CA
2025	AI for Medicine and Healthcare (**Invited Panel Talk**) *The Association for the Advancement of Artificial Intelligence (AAAI)*, Philadelphia, PA
2025	"From Models to Agents: Structuring Clinical AI through Reasoning, Generation, and Alignment" (**Invited Talk**) *Seoul National University*, Seoul, South Korea
2025	"From Models to Agents: Structuring Clinical AI through Reasoning, Generation, and Alignment" (**Keynote Talk**) *Workshop on Generative AI and Large Language Models for Neuroscience and Brain Imaging, Brain Informatics 2025*, Bari, Italy
2025	"From Models to Agents: Structuring Clinical AI through Reasoning, Generation, and Alignment" (**Invited Talk**) *Bioinformatics Research Seminar Series*, Western University, London, Canada

2026	"From Pixel to Mind: Generative AI for Smart Health and Patient World Model" **(Keynote Talk)** *GenAI4HealthBI: Generative AI for Smart Health and Biomedical Informatics, Chase 2026,* Pittsburgh, PA

Report of Technological and Other Scientific Innovations

Automatic pre-screening method for pneumothorax detection, 2017	Development of a software system for automatically detecting pneumothorax from CT images. The system has been validated internally by multiple radiologists. The innovation was filed as an **Invention Disclosure to Partners HealthCare** in 2017.
Inference of 3D chest CT images from 2D chest radiographs, 2019	Development of an invention to directly infer 3D chest CT images from 2D radiographs. The technique learns the mapping between the 2D image and the 3D spatial structure information of the corresponding CT image from the teaching dataset, achieving trans-dimensional 2D-3D mapping. The innovation was filed as an **Invention Disclosure to Partners HealthCare** in 2020.
Predicting Alzheimer's Disease by Hierarchical Graph Convolution from Positron Emission Tomography Imaging (PETNet), 2021	Developing a Graph Convolutional Network-based Positron Emission Tomography (PET) image analysis program PETNet. The program can automatically diagnose Alzheimer's Disease (AD) and Mild Cognitive Impairment (MCI) from the input PET imaging data. The program features a hierarchical graph inference algorithm for fine-grained analysis of large-scale voxel-wise data and a Neural Architecture Search (NAS) scheme for identifying the optimal graph representing the brain images. The invention has been filed as an **Innovation Disclosure to MGB**.
Brain Image Analysis by Graph-based Matched Signal Detection (MSD) Algorithm, 2021	Developing a Matched Signal Detection (MSD) algorithm for predicting whether the given signal (e.g., medical images) belongs to a specific graph (e.g., graph of disease / normal population). MSD algorithm identifies the subspace for graph-signal by eigenvectors of the Laplacian matrix of the graph, which provides a concise encoding of the graph structure. Graph Fourier transform (GFT) is applied to project the graph signals onto the identified subspace. Based on GFT-based processing of graph signals, we can then test signal models on graphs before with hypothesis tests. The invention has been filed and approved as an **Innovation Disclosure to MGB**.
Machine Learning Model for the Prediction of Early Discharge of Patient Underwent Transcatheter Aortic Valve Replacement (TAVR) Using Electronic Medical Record, 2023	Developing a machine learning algorithm to predict the early discharge (<36h) of patients who underwent transcatheter aortic valve replacement, given in-house electronic medical record (EMR) information. The prediction is based on a patient's hospital stay, including demographics, vital signs, medical history, related laboratory results, and medications. The invention has been filed for an **Innovation Disclosure to MGB**.
AUTOMATED DETECTION AND MANAGEMENT OF VAVLULAR HEART DISEASE	Developing an Aortic Stenosis Ensemble Risk Prediction (AS-ERP) Model. AS-ERP performs Aortic Stenosis patient outcome (length of stay and readmission) prediction based on the input Electronic Medical Records (EMR) data. The model utilizes an ensemble learning scheme consisting of three machine learning classifiers for patient outcome prediction. Internal validation performance meets the clinical acceptance criteria and is superior to the current

USING MACHINE LEARNING	risk score system developed by the Society of Thoracic Surgeons. **U.S. Patent No. 20240379239**
SYSTEM AND METHOD FOR SYNTHESIZING 3D VOLUMETRIC MEDICAL IMAGES USING A PROBABILISTIC-BASED MACHINE LEARNING MODEL	Developing a system with pose-conditioned diffusion models to synthesize high-resolution 3D CT-like volumetric medical images from standard 2D radiographs (e.g., X-rays). By estimating imaging pose, aligning and back-projecting radiographs into coarse 3D volumes, and refining them with a trained diffusion model, the approach produces diagnostically useful 3D reconstructions without additional radiation or CT hardware, enabling faster, lower-cost, and more accessible volumetric imaging for clinical tasks such as fracture assessment and surgical planning **U.S. Patent pending (PCT/US 2025048292)**
SYSTEM AND METHOD FOR TOKENIZATION OF ANATOMICAL STRUCTURES FOR MEDICAL IMAGING	Developing a machine-learning framework for anatomy-aware tokenization of 2D and 3D medical images, representing individual organs and anatomical structures as semantic disentangled token groups (SDTGs). By encoding images into holistic latent representations, separating organ-specific features, modeling inter- and intra-organ relationships, and adaptively reconstructing images from selected token groups, the approach enables controllable organ-level reconstruction, segmentation, retrieval, image editing, and multimodal analysis while improving interpretability, flexibility, and computational/storage efficiency. **U.S. Patent pending (Provisional Patent Application No. 63/874,042)**

Report of Scholarship

Peer-Reviewed Scholarship in print or other media:

Research Investigations *indicates co-first authorship, **indicates my mentee

1 J Sun, X Hu, X Huang, Y Liu, K Li, **X Li**, J Han, L Guo, T Liu, J Zhang. Inferring consistent functional interaction patterns from natural stimulus FMRI data. *NeuroImage*. 2012;61(4):987-99.

2 **X Li**, C Lim, K Li, L Guo, T Liu. Detecting brain state changes via fiber-centered functional connectivity analysis. *Neuroinformatics*. 2013;11(2):193.

3 X Zhang, L Guo, **X Li**, T Zhang, D Zhu, K Li, H Chen, J Lv, C Jin, Q Zhao, L Li, T Liu. Characterization of task-free and task-performance brain states via functional connectome patterns. *Medical Image Analysis*. 2013;17(8):1106.

4 **X Li**, D Zhu, X Jiang, C Jin, X Zhang, L Guo, J Zhang, X Hu, L Li, T Liu. Dynamic functional connectomics signatures for characterization and differentiation of PTSD patients. *Human Brain Mapping*. 2014;35(4):1761.

5 J Ou, Z Lian, L Xie, **X Li**, P Wang, Y Hao, D Zhu, R Jiang, Y Wang, Y Chen, J Zhang, T Liu. Atomic dynamic functional interaction patterns for characterization of ADHD. *Human Brain Mapping*. 2014;35(10):5262.

6 D Sabatinelli, D Frank, T Wanger, M Dhamala, B Adhikari, **X Li**. The timing and directional connectivity of human frontoparietal and ventral visual attention networks in emotional scene perception. *Neuroscience*. 2014;277:229.

7 J Zhang*, **X Li***, C Li, Z Lian, X Huang, G Zhong, D Zhu, K Li, C Jin, X Hu, J Han, L Guo, X Hu, L Li, T Liu. Inferring functional interaction and transition patterns via dynamic Bayesian variable partition models. *Human Brain Mapping*. 2014;35(7):3314.

8 X Zhang*, **X Li***, C Jin, H Chen, K Li, D Zhu, X Jiang, T Zhang, J Lv, X Hu, J Han, Q Zhao, L Guo, T Liu. Identifying and characterizing resting state networks in temporally dynamic functional connectomes. *Brain Topography*. 2014;27(6):747.

9 X Jiang, **X Li**, J Lv, T Zhang, S Zhang, L Guo, T Liu. Sparse representation of HCP grayordinate data reveals novel functional architecture of cerebral cortex. *Human Brain Mapping*. 2015;36(12):5301.
10 J Lv*, X Jiang*, **X Li***, D Zhu, H Chen, T Zhang, S Zhang, X Hu, J Han, H Huang, J Zhang, L Guo, T Liu. Sparse representation of whole-brain fMRI signals for identification of functional networks. *Medical Image Analysis*. 2015;20(1):112.
11 J Lv*, X Jiang*, **X Li***, D Zhu, S Zhang, S Zhao, H Chen, T Zhang, X Hu, J Han, J Ye, L Guo, T Liu. Holistic atlases of functional networks and interactions reveal reciprocal organizational architecture of cortical function. *IEEE Transactions on Biomedical Engineering*. 2015;62(4):1120.
12 J Lv, X Jiang, **X Li**, D Zhu, S Zhao, T Zhang, X Hu, J Han, L Guo, Z Li, C Coles, X Hu, T Liu. Assessing effects of prenatal alcohol exposure using group-wise sparse representation of fMRI data. *Psychiatry Research: Neuroimaging*. 2015;233(2):254.
13 M Makkie, S Zhao, X Jiang, J Lv, Y Zhao, B Ge, **X Li**, J Han, T Liu. HAFNI-enabled large-scale platform for neuroimaging informatics (HELPNI). *Brain Informatics*. 2015;2(4):225.
14 J Ou, L Xie, C Jin, **X Li**, D Zhu, R Jiang, Y Chen, J Zhang, L Li, T Liu. Characterizing and differentiating brain state dynamics via hidden Markov models. *Brain Topography*. 2015;28(5):666.
15 J Ou, L Xie, **X Li**, D Zhu, D Terry, A Puente, R Jiang, Y Chen, L Wang, D Shen, J Zhang, L Miller, T Liu. Atomic connectomics signatures for characterization and differentiation of mild cognitive impairment. *Brain Imaging and Behavior*. 2015;9(4):663.
16 Y Hou, T Xiao, S Zhang, X Jiang, **X Li**, X Hu, J Han, L Guo, L Miller, R Neupert, T Liu. Predicting Movie Trailer Viewer's "Like/Dislike" via Learned Shot Editing Patterns. *IEEE Transactions on Affective Computing*. 2016;7(1):29.
17 S Zhang*, **X Li***, J Lv, X Jiang, L Guo, T Liu. Characterizing and differentiating task-based and resting state fMRI signals via two-stage sparse representations. *Brain Imaging and Behavior*. 2016;10(1):21.
18 X Jiang, **X Li**, J Lv, S Zhao, S Zhang, W Zhang, T Zhang, J Han, L Guo, T Liu. Temporal dynamics assessment of spatial overlap pattern of functional brain networks reveals novel functional architecture of cerebral cortex. *IEEE Transactions on Biomedical Engineering*. 2016;65(6):1183.
19 B Ge, M Makkie, J Wang, S Zhao, X Jiang, **X Li**, J Lv, S Zhang, W Zhang, J Han, L Guo, T Liu. Signal sampling for efficient sparse representation of resting state FMRI data. *Brain Imaging and Behavior*. 2016;10:1206.
20 Y Li, H Chen, X Jiang, **X Li**, J Lv, M Li, H Peng, J Tsien, T Liu. Transcriptome Architecture of Adult Mouse Brain Revealed by Sparse Coding of Genome-Wide In Situ Hybridization Images. *Neuroinformatics*. 2017;15(3):285.
21 Y Li, H Chen, X Jiang, **X Li**, J Lv, H Peng, J Tsien, T Liu. Discover mouse gene coexpression landscapes using dictionary learning and sparse coding. *Brain Structure and Function*. 2017;222(9):4253.
22 J Yuan, **X Li**, J Zhang, L Luo, Q Dong, J Lv, Y Zhao, X Jiang, S Zhang, W Zhang, T Liu. Spatio-temporal modeling of connectome-scale brain network interactions via time-evolving graphs. *NeuroImage*. 2017;180:350.
23 B Ge, **X Li**, X Jiang, Y Sun, T Liu. A Dictionary Learning Approach for Signal Sampling in Task-based fMRI for Reduction of Big Data. *Frontiers in Neuroinformatics*. 2018;12.
24 M Makkie*, **X Li***, S Quinn, B Lin, J Ye, G Mon, T Liu. A Distributed Computing Platform for fMRI Big Data Analytics. *IEEE Transactions on Big Data*. 2018;5(2):109.
25 J Thrall, **X Li**, Q Li, C Cruz, S Do, K Dreyer, J Brink. Artificial Intelligence and Machine Learning in Radiology: Opportunities, Challenges, Pitfalls, and Criteria for Success. *Journal of the American College of Radiology*. 2018;15(3):504.
26 W Zhang, J Lv, **X Li**, D Zhu, X Jiang, S Zhang, Y Zhao, L Guo, J Ye, D Hu, T Liu. Experimental Comparisons of Sparse Dictionary Learning and Independent Component Analysis for Brain Network Inference from fMRI Data. *IEEE Transactions on Biomedical Engineering*. 2018;66(1):289.

27 Z Guo**, **X Li***, H Huang, N Guo, Q Li. Deep Learning-based Image Segmentation on Multi-modal Medical Imaging. *IEEE Transactions on Radiation and Plasma Medical Sciences*. 2019;3(2):162.
28 **X Li**, N Guo, Q Li. Functional Neuroimaging in the New Era of Big Data. *Genomics Proteomics and Bioinformatics*. 2019;17(4):393.
29 H Wang, K Xie, L Xie, **X Li**, M Li, C Lyu, H Chen, Y Chen, X Liu, J Tsien, T Liu. Functional Brain Connectivity Revealed by Sparse Coding of Large-Scale Local Field Potential Dynamics. *Brain Topography*. 2019;32(2):255.
30 Y Zhao*, **X Li***, H Huang, W Zhang, S Zhao, M Makkie, M Zhang, Q Li, T Liu. 4D Modeling of fMRI Data via Spatio-Temporal Convolutional Neural Networks (ST-CNN). *IEEE Transactions on Cognitive and Developmental Systems*. 2019;12(3):451.
31 **X Li**, JH Thrall, SR Digumarthy, MK Kalra, PV Pandharipande, B Zhang, C Nitiwarangkul, R Singh, RD Khera, Q Li. Deep learning-enabled system for rapid pneumothorax screening on chest CT. *European Journal of Radiology*. 2019;120:108692.
32 S Jeong*, **X Li***, J Yang, Q Li, V Tarokh. Sparse Representation-Based Denoising for High-Resolution Brain Activation and Functional Connectivity Modeling: A Task fMRI Study. *IEEE Access*. 2020;8:36728.
33 M Zhang**, **X Li***, M Xu, Q Li. Automated Semantic Segmentation of Red Blood Cells for Sickle Cell Disease. *IEEE Journal of Biomedical and Health Informatics*. 2020;24:3095.
34 P Wang, X Jiang, H Chen, S Zhang, **X Li**, Q Cao, L Sun, L Liu, B Yang, Y Wang. Assessing Fine-Granularity Structural and Functional Connectivity in Children with Attention Deficit Hyperactivity Disorder. *Frontiers in Human Neuroscience*. 2020;14:481.
35 X Wang, L Zhang, X Yang, L Tang, J Zhao, G Chen, **X Li**, S Yan, S Li, Y Yang, Y Kang, Q Li, N Wu. Deep Learning Combined with Radiomics May Optimize the Prediction in Differentiating High-Grade Lung Adenocarcinomas in Ground Glass Opacity Lesions on CT Scans. *European Journal of Radiology*. 2020;129:109150.
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Narrative Report

I am an Assistant Professor at Massachusetts General Hospital and Harvard Medical School, Department of Radiology, and an Affiliate Faculty Member at the Kempner Institute for Natural and Artificial Intelligence of Harvard University. My main work is the research investigation into Artificial General Intelligence (AGI) for healthcare, aiming at building AI systems that learn broadly across many types of

medical data and can be adapted to many clinical tasks, with focus on human/expert–machine alignment, knowledge-grounded trustworthiness, and privacy-preserving computing. I am also responsible for developing computer-assisted solutions for clinical applications, leading several projects that have been translated to patents, physician-training platforms, and hospital-deployed systems for screening and risk prediction. I have authored over 170 peer-reviewed articles that have been cited more than 17,000 times (H-index 60). My standing in the field is reflected in my service as area chair for NeurIPS, AAAI, KDD, MICCAI, and ACM Multimedia; as associate or guest editor for journals including PNAS, IEEE TMI and IEEE TAI; and grant reviewer for NIH study sections, the Dutch Research Council (NWO), and the Society of Nuclear Medicine and Molecular Imaging (SNMMI).;

Research Investigation

My research focuses on developing generalized, scalable, and reliable AGI solutions for medical data analytics. I articulated this vision in the **Featured Article** of *IEEE Reviews in Biomedical Engineering* [64], and have pursued it along three lines of research that complement each other.

First, I develop vertical, domain-specific foundation models in medicine by combining clinical domain priors with AI algorithm design. As co-corresponding author, I led the development of **BiomedGPT**, a multi-modal biomedical foundation model featured in *Nature Medicine* [60], and, extending this paradigm to neuroimaging, a general-purpose foundation model for functional MRI analysis published in *Nature Biomedical Engineering* [83]. I further developed the large language model (LLM) **Radiology-GPT** for radiology report understanding and summarization **[3]**, as well as RadOncGPT for radiation oncology text processing, GP-GPT for gene–phenotype data mapping, and MA-SAM for modality-agnostic 3D medical image segmentation. I was awarded the Google Research Scholar Program to develop LLM+EHR models for late-life depression management, and my work on augmenting LLMs with external knowledge bases (MKRAG) was selected by AMIA 2024 for the Distinguished Paper Award.

Second, I develop generative AI methods for healthcare data. I have built GAN-based and diffusion-based models for synthesizing medical image data and for temporal trajectory prediction. I am especially interested in reducing imaging cost and radiation exposure via generative AI, including models for X-ray-to-CT synthesis (supported by the American Foundation for Surgery of the Hand) and ECG-to-echo synthesis.

Third, I design AI algorithms that better accommodate the unique challenges and needs of clinical scenarios, including federated learning frameworks to overcome institutional data barriers (Nature Medicine), advanced fusion mechanisms for multi-modal medical images (IEEE TRPMS Best Paper Award, 2022), and advanced representation methods for medical images (IEEE ISBI Best Student Paper / Best Paper Awards, 2011, 2013, 2020). Most recently, I am extending medical foundation models toward clinically grounded, interpretable agentic systems, including an agentic AI for radiology with localized reasoning, published in npj Digital Medicine **[4]**.

My research mainly targets healthcare problems in cardiovascular disease (heart failure and valvular disease), mental disorders (depression, PTSD, and OCD), and Alzheimer's disease, utilizing a wide spectrum of medical data modalities including imaging, clinical text, electronic health records (EHR), physiological waveforms, and multi-modal fusion.

My research program is supported by independent, peer-reviewed federal funding. I currently lead an NIH R01 as Multiple-PI (UniBrain, an end-to-end unified toolkit for comprehensive neuroimaging analysis; $2.5M direct costs) and serve as the MGH subcontract PI on a second NIH R01 developing visually grounded PET report generation via federated learning ($2.3M direct costs). I previously held an NIH R03 as PI on multi-modal imaging biomarkers for early prediction of MCI-to-AD conversion.

Technological Innovation

At Massachusetts General Hospital, I have collaborated with physicians, radiologists, and system engineers to deliver novel solutions for medical data. My innovations are protected by one issued US patent and two pending applications: an issued patent (US Patent No. 20240379239) for an Aortic Stenosis Ensemble Risk Prediction (AS-ERP) model that uses EHR data to predict patient outcomes (length of stay and readmission); a pending patent on a diffusion-based method that synthesizes 3D CT volumes from 2D radiographs, which achieved clinically acceptable performance through a rigorous human reader study; and a pending patent on anatomy-aware tokenization of medical images for controllable, organ-level analysis.

Service to the Community

I serve on the editorial boards and as area chair for several international journals and conferences, including IEEE TMI, IEEE TAI, NeurIPS, AAAI, KDD, ACM MM, and MICCAI. I have also served as a grant review panelist for several NIH study sections. To promote the importance and advancement of multi-modal, multi-scale medical image analysis and to facilitate interaction between clinical and data science experts, I founded and chaired the International Workshop on Multiscale Multimodal Medical Imaging (MMMI) since 2019, the International Workshop on Embodied AI and Robotics for Healthcare (EARTH) 2024, the SAGES Critical View of Safety Challenge 2024, and the NeurIPS Workshop on Advancements in Medical Foundation Models 2024. These initiatives facilitate collaboration between clinicians and AI researchers while advancing the field of multi-modal AGI in healthcare.

Teaching and Educational Activities

I currently serve on the thesis committees of three PhD students, and I have mentored six research fellows and many visiting students at MGH. Through mentorship and close collaboration on research projects, most of my fellows and students have produced more than one publication in scientific journals or conferences; several have led first-author papers in leading venues, including echocardiography foundation models (EchoFM and MediViSTA) in IEEE Transactions on Medical Imaging and IEEE Journal of Biomedical and Health Informatics. In addition, I have given multiple lectures on healthcare AI, computational modeling, and solution development in radiology. Since 2021, I have organized the “AI x Med” bi-weekly seminar, a lecture series on the latest developments in AI applications in medicine, featuring researchers within the hospital and invited speakers from external institutions.